

PermRepo User Guide

1. What is PermRepo?

PermRepo is a long-term backup system for GitHub repositories.

PermRepo allows saving the contents of a GitHub repository so that backup data is not stored on a single centralized server. Backup data is encrypted on the user side and stored on the Arweave network using Turbo upload infrastructure.

PermRepo also uses the Base blockchain to maintain a permanent on-chain information object for each repository — On-chain Descriptor.

Simplified:

GitHub → PermRepo → local encryption → Turbo → Arweave

and in parallel:

GitHub repository → On-chain Descriptor → Base blockchain

The PermRepo server is not a place where the user's Master Key is stored.

2. What is an On-chain Descriptor?

An On-chain Descriptor is a permanent blockchain object that identifies a PermRepo repository and records its current backup state.

An On-chain Descriptor is not:

- the GitHub repository itself;
- a backup file;
- Master Key;
- password;
- access key;
- encryption key;
- legal copyright certificate;
- tradable asset.

An On-chain Descriptor serves as a blockchain-verifiable repository backup identification and state anchor.

This allows storing information on the blockchain, such as:

- repository identification;
- number of backups;
- last manifest URI;
- last Merkle root value;
- backup sequence information.

Thus, the backup history does not depend solely on the PermRepo server database.

3. Why is a blockchain used?

The blockchain in the PermRepo system is not used for storing the backup files themselves.

Storing large backup files on the blockchain is not practical.

The blockchain is used as a publicly verifiable information layer.

The PermRepo On-chain Descriptor can point to the latest backup manifest, but the manifest itself and the backup data reside on Arweave.

This creates two distinct layers:

Arweave

Stores:

- encrypted backup archives;
- manifests;
- backup history and related data.

Base blockchain

Stores:

- On-chain Descriptor;
- backup count;
- last manifest URI;
- Merkle root;
- blockchain state associated with the backup object.

4. What is required to use PermRepo?

To perform a backup, the user needs several independent elements.

4.1. GitHub account

The GitHub account determines:

- who the user is;
- which repositories are available to them;
- which repository is allowed to be backed up.

PermRepo verifies whether the user has access rights to the specific GitHub repository.

4.2. Cryptocurrency wallet

An EVM-compatible wallet is required, such as MetaMask.

The wallet is used to:

- create and manage the On-chain Descriptor;
- prove control over the Descriptor;
- sign blockchain actions;
- migrate the On-chain Descriptor to another wallet owned by the user.

PermRepo does not acquire the user's private wallet key.

4.3. Active PermRepo subscription

An active PermRepo subscription is required for backup functionality.

The subscription is separate from Turbo Credits.

The PermRepo subscription is tied to the GitHub account, not to a specific wallet.

This means that changing wallets in itself does not create a need to transfer the subscription.

In the current PermRepo configuration, the subscription price is 2.5 USDC per month.

4.4. Turbo Credits

Turbo Credits are required to upload backup data.

Turbo Credits are not PermRepo credits.

Turbo Credits are a payment method for the Turbo upload service used to pay for data storage on the Arweave network. AR.IO documentation describes Turbo Credits as an upload payment method whose value corresponds to Arweave data storage capacity.

Purchasing Turbo Credits does not occur on the PermRepo platform.

The user purchases them outside of PermRepo, for example:

- AR.IO Console: console.ar.io
- ArDrive: ardrive.io

AR.IO documentation currently states that the Console can be used to purchase Turbo Credits with fiat or supported crypto assets, while ArDrive provides its own Turbo/credits purchase flow.

PermRepo:

- does not sell Turbo Credits;
- does not charge a fee for purchasing Turbo Credits;
- does not manage the user's Turbo Credits balance;
- does not store the user's Turbo Credits;
- does not use its payment account to fund user backups.

During backup upload, Turbo uses the credits available in the user's Turbo account.

5. Subscription and Turbo Credits are two different things

This distinction is important.

	PermRepo subscription	Turbo Credits
What it is for	Using the PermRepo service	Paying for Arweave data uploads
Where to purchase	PermRepo	AR.IO Console / ArDrive
Associated with	GitHub account	Turbo user wallet/account
Does PermRepo sell it?	Yes	No
Required for backup?	Yes	Yes
Does changing wallets change the subscription?	No	Turbo account terms are separate

Therefore, an active PermRepo subscription alone does not mean that backup uploading will be possible.

A sufficient Turbo Credits balance is also required to perform a backup.

6. First Time Use

A typical sequence for first-time use is as follows:

1. Connect GitHub.
2. Connect cryptocurrency wallet.
3. Make sure the wallet operates on the Base network.
4. Purchase a PermRepo subscription.
5. Purchase sufficient Turbo Credits outside PermRepo.
6. Select a GitHub repository.
7. Create an On-chain Descriptor for this repository.
8. Perform the first backup.

7. Creating an On-chain Descriptor

An individual On-chain Descriptor is created for each repository.

Its creation links:

GitHub repository ↔ On-chain Descriptor ↔ user wallet

After creation, the specific repository can be identified in the blockchain object and its backup state tracked.

An On-chain Descriptor does not need to be created anew for each backup.

The same Descriptor is used for one repository throughout its entire backup history.

8. First Backup

The first backup differs from subsequent ones in that a Master Key is created.

During the first backup:

1. PermRepo receives information about the GitHub repository.
2. Repository files are identified.
3. Backup is prepared in the browser.
4. A random 32-byte Master Key is generated in the browser.
5. The Master Key is shown to the user.
6. Files are placed into a ZIP archive.
7. The ZIP is encrypted using AES-GCM.
8. The encrypted archive is uploaded to Turbo.
9. A manifest is created.
10. The manifest is uploaded to Turbo.
11. Backup information is recorded on the blockchain.

9. What is a Permanent Web Backup Restore Platform?

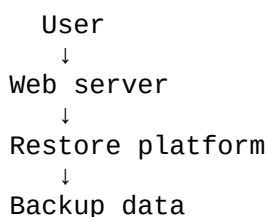
The PermRepo backup restore platform is not designed as a regular centralized webpage residing on a PermRepo server.

It is published on the Arweave network as a permanent web application.

This means the restore platform itself is preserved alongside permanent web infrastructure principles, rather than relying on a PermRepo server remaining operational in the future.

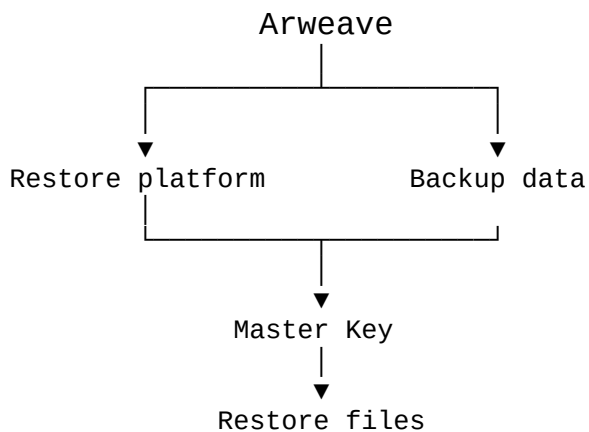
Why is this important?

In a standard web app, the situation might look like this:



If the web server stops working, the restore platform may also become unavailable.

PermRepo uses a different principle:



Therefore, the availability of the backup restore platform does not depend on the health of the PermRepo backend server.

10. What does Permanent Web mean?

The Permanent Web principle means that web content and applications can be stored on a permanent decentralized data network rather than just on a single traditional web server.

In the case of PermRepo, this is especially significant because the goal of a backup system is not just to save data today.

The goal is to ensure access to this data in the future.

Therefore, not only are backup files and manifests preserved, but also the application itself required to restore these backups.

This reduces reliance on:

- PermRepo server;
- PermRepo backend uptime;
- specific hosting service;
- operation of a centralized backup restore platform.

11. What does this mean for the user?

The backup restore infrastructure is built to be accessible directly from the permanent web layer.

Thus, the PermRepo backend is mainly needed during backup creation and service operation, not for the existence of the permanent restore platform.

This is a fundamental distinction:

Backup Creation Side:

GitHub
↓
PermRepo backend/frontend
↓
Turbo
↓
Arweave
↓
Base

Backup Restoration Side:

Arweave
↓
Permanent Web Restore Platform
↓
Arweave Backup Data
↓
Master Key
↓
Restored Files

12. Does the PermRepo server need to be running to restore a backup?

The operation of the backup restore platform does not depend on the PermRepo server being operational.

This is one of the main benefits of Permanent Web architecture.

If in the future:

- PermRepo backend server is shut down;
- PermRepo server encounters a technical failure;
- PermRepo company no longer maintains the backend;

this in itself must not mean that the permanent restore platform disappears.

The restore platform and backup data are published on Arweave's permanent layer!

13. Master Key

The Master Key is one of the most critical responsibilities of a PermRepo user.

It is generated in the user's browser, not on the PermRepo server.

Master Key:

- is not sent to the PermRepo server;
- is not stored in Redis;
- is not placed on the blockchain;
- is not published on Arweave;
- is used to encrypt backup data.

The user must store the Master Key themselves.

The recommended location is a password manager or another secure, user-controlled vault.

Important!

PermRepo cannot recover a lost Master Key.

If the Master Key is lost and no copy exists, decrypting the encrypted backup data is impossible.

14. Master Key and Cryptocurrency Wallet are not the same thing

These two items must not be confused.

Cryptocurrency Wallet

Cryptocurrency Wallet provides:

- blockchain identity;
- control over On-chain Descriptor;
- signing blockchain transactions;
- Descriptor migration;
- authorization of new backup actions.

Master Key

Master Key provides:

- access to decrypt encrypted backup data;
- ability to restore backup content when backup data is available.

Therefore:

Cryptocurrency Wallet $\neq$ Master Key

and

Master Key $\neq$ Cryptocurrency Wallet

15. What happens if access to the wallet is lost?

Losing wallet access and losing the Master Key are two different situations.

If a user loses access to their cryptocurrency wallet, it means they can no longer use that wallet for blockchain actions.

Loss of crypto wallet access affects:

- new backups;
- On-chain Descriptor management;
- migration from the old wallet.

However, it does not automatically mean that previously created backup data is lost.

If the user has saved their Master Key and the backup data is still available on Arweave, restoring existing backups does not depend on the user still controlling the old wallet.

Therefore, it is important to separately protect:

- cryptocurrency wallet access;
- Master Key.

16. On-chain Descriptor Migration

The PermRepo On-chain Descriptor is not intended for sale or free transfer between users.

It is migratable so that a user can change their wallet.

This is important because users may:

- change wallets;
- switch to a hardware wallet;
- change wallet infrastructure;
- update their security model.

Migration is not a sale. Migration is a user-controlled transfer from one wallet they control to another wallet they control.

17. Can an On-chain Descriptor be sold?

No, the PermRepo On-chain Descriptor is not designed as a tradable asset.

Standard ERC-721 transfer functions are blocked.

This means the Descriptor is not intended for:

- sale;
- purchase;
- exchanges;
- marketplaces;
- speculation.

The only intended owner-change mechanism is migration.

18. Subsequent Backups

After the first backup, the user does not need to generate a new Master Key for each backup.

During subsequent backups:

1. User selects the same repository.
2. PermRepo verifies subscription.
3. On-chain Descriptor is checked.
4. Current GitHub state is compared with previous backup history.
5. Changed files are identified.
6. User enters their existing Master Key.
7. Changed files are placed in a ZIP.
8. ZIP is encrypted.
9. Encrypted ZIP is uploaded to Turbo.
10. New manifest is created.
11. Manifest is uploaded to Turbo.
12. New backup is registered on the On-chain Descriptor.

19. Incremental Backup

PermRepo is not designed to blindly upload the entire repository every time.

The system compares current GitHub file states with previous backup information.

If a file has not changed, it does not need to be uploaded again.

If a file has changed, it is included in the new backup archive.

Incremental Backup reduces:

- uploaded data volume;
- Turbo Credits consumption;
- backup creation time.

The manifest preserves information linking the current backup to previous backup history.

20. What is a Manifest?

A manifest is a structured JSON document describing a specific backup.

The manifest is not the backup archive itself.

The manifest may contain information about:

- backup number;
- encrypted archive identifier;
- archive location;
- encryption IV;
- changed files;
- file hash values;
- previous manifests;
- backup history.

Thus, the manifest acts as a description and link point for a specific backup.

21. What is a Merkle Root?

During the PermRepo backup process, a Merkle root value related to the backup file state is calculated.

The Merkle root is recorded on the blockchain alongside backup information.

The Merkle root provides publicly verifiable information regarding the file state of a specific backup.

Storing all file content on the blockchain is unnecessary. Instead, a compact cryptographic value is stored on the blockchain, which can be used in the backup integrity verification process.

22. How are files protected?

Backup data is encrypted in the user's browser before upload.

Process is:

GitHub files → ZIP → AES-GCM encryption → Turbo → Arweave

Therefore, the PermRepo server does not need to store unencrypted backup archives.

The Master Key remains under user control.

23. Does the PermRepo server store my backup files?

The PermRepo server is not a permanent backup data repository.

The server is mainly used for:

- GitHub authentication;
- GitHub API operations;
- session management;
- backup process state management;
- preparing necessary authorizations.

Permanent backup data resides on Arweave/Turbo infrastructure.

Blockchain maintains the On-chain Descriptor and backup state information.

24. Role of Turbo

Turbo is upload and payment infrastructure used to deliver data to Arweave storage.

Users do not need to understand all technical details of Turbo.

Practically, users only need to know:

- backup data is stored on Arweave;
- PermRepo uses Turbo for uploads;
- Turbo Credits are used to pay for uploads;
- Turbo Credits must be purchased outside PermRepo.

AR.IO documentation describes Turbo as Arweave data upload and payment infrastructure.

25. Where to buy Turbo Credits?

Turbo Credits do not need to be purchased on PermRepo.

Users can use:

AR.IO Console console.ar.io

AR.IO documentation indicates Console top-up interface for purchasing Turbo Credits:

console.ar.io/topup

ArDrive ardrive.io

ArDrive also provides Turbo Credits purchasing options.

After purchasing credits, they are used for Turbo uploads.

26. How many Turbo Credits are needed?

Turbo Credits consumption depends on the volume of data being uploaded and the specific Turbo pricing model.

Therefore, PermRepo users should not view Turbo Credits as a fixed "one backup = one credit" system.

Larger backups mean larger data volume and higher storage costs.

That is why incremental backups are important.

Turbo calculates upload costs itself and automatically consumes available credits.

27. What happens if Turbo Credits are insufficient?

If there are insufficient credits in the Turbo account, the backup upload cannot be completed.

In this case:

- backup process is not considered successfully finished;
- user must top up Turbo Credits;
- backup process can then be continued or repeated according to PermRepo process state.

This is not the same as PermRepo subscription expiration.

Subscription and Turbo Credits are independent.

28. Security of the Backup Process

PermRepo uses multiple distinct protection layers.

GitHub authorization

PermRepo verifies whether the user has access rights to the specific repository.

Crypto wallet authorization

The user's crypto wallet is used for blockchain actions.

EIP-712

Structured signatures used for blockchain authorizations.

Nonce

Nonce mechanisms used for blockchain actions to prevent signature replay attacks.

Deadline

Signed authorizations have expiration times.

AES-GCM

Backup ZIP encrypted before upload.

Master Key

Encryption key stays under user control.

29. What happens if the backup process is interrupted?

The backup process consists of several stages:

1. backup preparation;
2. ZIP creation;
3. ZIP encryption;
4. ZIP upload;
5. manifest creation;
6. manifest upload;
7. blockchain authorization;
8. addBackup transaction;
9. process completion.

If interrupted, PermRepo saves the necessary backup job state so the process can be recovered rather than starting from scratch.

This is especially crucial for the first backup so the Master Key is not regenerated due to an upload failure.

30. What happens if the browser closes?

If interrupted unexpectedly, PermRepo uses saved backup process state to determine where it stopped.

If the blockchain transaction was already sent, the system can await confirmation and complete backup process registration.

This prevents scenarios where the blockchain transaction succeeds, but the user closed the browser before UI showed completion.

31. Backup Restoration

Backup data is stored encrypted.

To decrypt an existing backup, the user's Master Key is required.

Key principle:

crypto wallet controls blockchain actions, while Master Key controls access to encrypted backup data.

Losing access to the crypto wallet does not mean backup data is lost.

If the user has:

- saved Master Key;
- access to the backup;
- backup manifest data;

restoring existing backups does not depend on controlling the old wallet.

32. What happens if Master Key is lost?

This is much more serious.

If Master Key is lost and no backup copy exists, PermRepo cannot recover it for the user.

PermRepo server does not have Master Key.

Blockchain does not have Master Key.

Arweave does not have Master Key.

Master Key must be treated as the user's personal master secret.

Store in at least one secure password manager entry and, if necessary, in another secure backup location.

33. Can PermRepo see my Master Key?

No, Master Key is generated or entered in the user's browser.

It is not transmitted to the PermRepo server.

This is one of the primary security principles of PermRepo.

34. Can PermRepo decrypt my backups?

Not without the Master Key.

Backup archives are encrypted prior to upload.

Master Key remains user-side.

Thus, storing backups on Arweave does not make contents publicly readable.

35. What happens if a GitHub repository is deleted?

PermRepo backup archives and manifests for existing backups do not depend on the GitHub repository continuing to exist.

GitHub is the original data source.

Arweave is the permanent backup storage layer.

Blockchain maintains backup identification and state information.

Deleting the GitHub repository does not delete already created backup data.

36. Are backup data public?

Arweave is a public decentralized network.

However, PermRepo backup archives are encrypted before upload.

The encrypted data unit, manifest, and blockchain info are public, but file contents cannot be read without the Master Key.

The user should remember that encryption protects file content but does not automatically make all blockchain or Arweave metadata private.

37. What is Backup History?

Each successful backup links to previous backup history.

Manifests preserve sequence tracking information.

Blockchain updates backup count and latest manifest info.

Thus, it can be seen that a specific repository has had multiple consecutive backups.

38. What does the user see in PermRepo?

Main interface displays information about the repository, such as:

- GitHub repository name;
- On-chain Descriptor ID;
- Backup count;
- Latest manifest;
- Latest Merkle root;
- Subscription status.

During backup process, the user also sees:

- file count;
- data volume;
- ZIP creation;
- encryption process;
- upload process;
- manifest creation;
- blockchain signing;
- backup completion status.

39. What does the PermRepo server NOT do?

PermRepo server:

- does not store user Master Key;
- does not store crypto wallet private keys;
- does not process Turbo payments for users;
- does not buy Turbo Credits on behalf of users;
- is not permanent backup data storage;
- does not mint On-chain Descriptors on behalf of users.

Blockchain actions requiring user crypto wallet signature are performed with the user's own wallet.

40. What happens if user changes crypto wallet?

Changing crypto wallet is accommodated in PermRepo's operating model.

Instead of creating a new repository identity and losing previous blockchain history, the user can use On-chain Descriptor migration.

The goal of migration is to preserve the existing repository blockchain object and transfer its control to a new crypto wallet owned by the user.

This is especially important for users who regularly change or rotate their crypto wallets.

41. What must the user store?

The user must store at least:

1. GitHub account access

Without a GitHub account, PermRepo cannot verify repository access rights.

2. Crypto wallet access

Crypto wallet is required for blockchain operations and On-chain Descriptor control.

3. Master Key

Master Key is required to decrypt encrypted backup data.

4. Turbo Credits

The user must ensure a sufficient Turbo Credits balance if they wish to perform new backups.

42. What MUST NOT be done with the Master Key?

The Master Key should not be:

- published on GitHub;
- put in repository code;
- sent via Discord or Slack;
- put in GitHub Issues;
- stored in unencrypted public files;
- sent to PermRepo server;
- published in a blockchain transaction;
- published in an Arweave manifest.

Master Key must be treated as a secret encryption key.

43. What to do if backup fails?

First check:

GitHub

- whether GitHub session is active;
- whether repository is still accessible.

Wallet

- whether crypto wallet is connected;
- whether correct network is selected;
- whether connected crypto wallet controls the corresponding On-chain Descriptor.

Subscription

- whether PermRepo subscription is active.

Turbo Credits

- whether Turbo account has sufficient credits for backup upload.

Master Key

- whether entered Master Key is correct.

Blockchain

- whether user rejected signature or transaction request.

44. If "Master Key is required" appears

This means PermRepo cannot continue backup without Master Key.

In second and subsequent backups, the exact same Master Key used for the first backup must be used.

If the entered key is incorrect, backup cannot proceed properly.

45. If wallet owner mismatch appears

This means connected crypto wallet does not control the specific repository's On-chain Descriptor.

Check:

- whether correct crypto wallet is connected;
- whether correct GitHub repo is selected;
- whether On-chain Descriptor was not migrated to another crypto wallet.

46. If insufficient Turbo Credits

PermRepo cannot replace Turbo payment with its own payment in this situation.

The user must top up Turbo Credits in their Turbo account.

Use:

AR.IO Console console.ar.io

or

ArDrive ardrive.io

After topping up credits, backup process can continue.

Turbo documentation indicates credits are automatically used during upload.

47. Are Turbo Credits the same as AR?

Turbo Credits are a Turbo payment method for uploads.

They reflect Arweave storage capacity and are used in the Turbo service.

Turbo Credits are not intended as a regular tradable crypto asset. AR.IO documentation indicates Turbo Credits are not intended for secondary trading or exchange for other cryptocurrencies.

48. Do Turbo Credits belong to PermRepo?

No, Turbo Credits do not belong to PermRepo.

Turbo Credits are a resource of the user's Turbo account.

PermRepo only uses the Turbo upload service.

Therefore PermRepo cannot refund, transfer or administer user Turbo Credits balance.

Turbo documentation also provides for checking credit balances and in certain cases sharing credits with other wallets.

49. Privacy

PermRepo uses several distinct information layers:

GitHub

Identifies user and repository access rights.

PermRepo server

Processes session and backup process state.

Blockchain

Stores On-chain Descriptor and publicly verifiable backup state info.

Arweave

Stores permanent backup data and manifests.

Master Key

Remains under user control and is used to protect encrypted data.

50. Main Security Principle

PermRepo security model is based on separating distinct control functions.

GitHub → determines which repository user has access to.

Crypto wallet → provides blockchain control and signing.

On-chain Descriptor → links repository to blockchain backup history.

Turbo → provides data upload infrastructure.

Arweave → provides permanent data storage.

Master Key → protects backup data content.

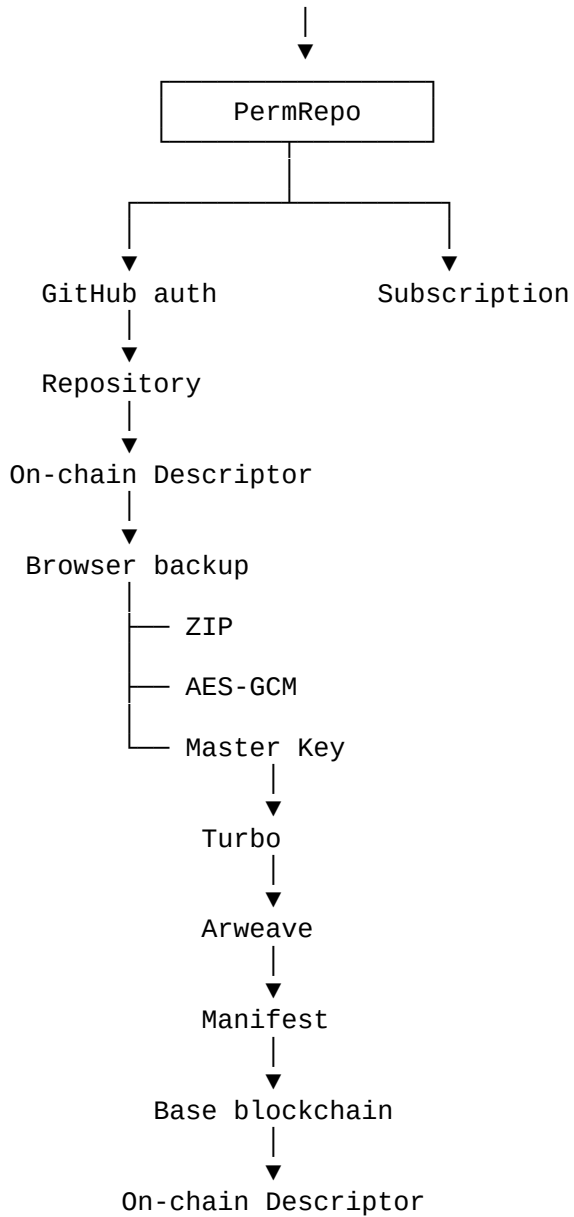
PermRepo subscription → determines access to PermRepo service.

These elements are not mutually interchangeable.

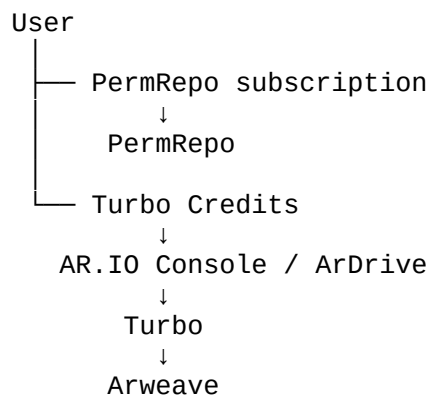
51. Summary Scheme

Full process in one diagram:

GITHUB



Turbo Credits are outside of this PermRepo platform payment flow:



52. Key Takeaways for the User

There are four key principles in using PermRepo.

1. Save Master Key

Without it, decrypting encrypted backups is impossible.

2. Protect your crypto wallet

Crypto wallet controls blockchain actions and On-chain Descriptor.

3. Maintain active PermRepo subscription

Subscription is required for PermRepo backup functionality.

4. Maintain sufficient Turbo Credits balance

Turbo Credits are required for backup data upload.

Turbo Credits are purchased outside PermRepo:

- AR.IO Console console.ar.io
- ArDrive ardrive.io

53. Essence of PermRepo in One Sentence

PermRepo allows users to save encrypted GitHub repository backups to Arweave using Turbo infrastructure, while utilizing Base blockchain as a permanent On-chain Descriptor and backup state verification layer.

User controls two vital elements:

crypto wallet — for blockchain control

and

Master Key — for backup data decryption.

PermRepo subscription and Turbo Credits are separate service elements, and **Turbo Credits are purchased outside PermRepo — AR.IO Console or ArDrive.**

Quick Summary Reminder

- **PermRepo subscription** → PermRepo service
- **Turbo Credits** → backup data upload
- **Crypto Wallet** → blockchain control
- **On-chain Descriptor** → repository backup state blockchain anchor
- **Master Key** → backup data decryption
- **Arweave** → permanent backup data storage
- **Turbo** → upload infrastructure